# Chapter 1: Foundations of Molecular Biology

Cells are the fundamental structural and functional units of all known living organisms.

The cell theory, developed in the mid-nineteenth century, establishes that all living matter is composed of one or more cells, and all cells arise from pre-existing cells through division.

Eukaryotic cells contain membrane-bound organelles such as mitochondria, the endoplasmic reticulum, and the nucleus, which compartmentalize biochemical pathways and cellular respiration.

# Chapter 2: Bioenergetics and ATP Synthesis

Cellular respiration converts biochemical energy from nutrients into adenosine triphosphate.

Glycolysis occurs in the cytosol without requiring molecular oxygen, yielding two pyruvate molecules, two net ATP molecules, and two NADH reducing equivalents.

The citric acid cycle and oxidative phosphorylation take place within the inner mitochondrial membrane, driving the chemiosmotic generation of proton gradients.

# Chapter 3: Genetics, Transcription and Translation

Deoxyribonucleic acid stores biological genetic instructions organized as codons.

During transcription, RNA polymerase synthesizes messenger RNA from the template DNA strand.

Ribosomes translate the nucleotide triplets into specific amino acid polypeptide sequences, forming structural and catalytic proteins essential for organismal homeostasis.